

Sample Question Paper I

Subject: BCAC 0026 – Analysis of Algorithm and Data Structures

Time: 3 Hours

Max Marks: 50

Section A (4 Marks $\times$ 5 = 20 Marks)

Answer any five questions. Each question carries 4 marks.

1. Explain the Algorithm for insert element in a Linear Queue.
 2. Explain the working of **Merge Sort** with an example. What is its time complexity?
 3. Write a recursive function in C or Java to calculate the **factorial** of a number. Also mention advantages and disadvantages of recursion.
 4. What is a **Binary Search Tree (BST)**? Construct a BST using the elements: 50, 30, 20, 40, 70, 60, 80 and show **inorder traversal**.
 5. **Scenario-Based Question:**
Your college library management system stores student book issue records. To display all issued records quickly and in a sorted manner by issue date, which sorting algorithm would you choose among **Bubble Sort, Merge Sort, and Quick Sort**? Justify your choice with reasons.
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Section B (3 Marks $\times$ 5 = 15 Marks)

Answer any five questions. Each question carries 3 marks.

1. What is the **best-case and worst-case** time complexity of **Bubble Sort**?
 2. Differentiate between **Stack** and **Queue** with examples.
 3. Write the recursive definition of the **Fibonacci sequence**.
 4. What are the three types of **binary tree traversals**? Briefly explain each.
 5. **Scenario-Based Question:**
You are working on a browser history feature that allows users to move backward and forward through pages. Which type of **linked list** would be best suited for this task? Justify your answer.
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Section C (5 Marks $\times$ 3 = 15 Marks)

Answer any three questions. Each question carries 5 marks.

1. Write a program (in C, C++, or Java) to **implement Quick Sort**. Explain how pivoting works in your example.
2. Compare and contrast **Merge Sort, Quick Sort, and Insertion Sort** in terms of their algorithm strategy, complexity, and practical use.
3. **Scenario-Based Question:**
Suppose you are building an auto-suggestion feature for a search bar where new suggestions are continuously added and need to be displayed in sorted order. Which **tree-based structure or algorithm** would you use to maintain order efficiently while supporting frequent insertions and searches? Explain your reasoning.

Sample Question Paper – II

Subject: BCAC 0026 – Analysis of Algorithm and Data Structures**

Time: 3 Hours

Max Marks: 50

Section A (4 Marks $\times$ 5 = 20 Marks)

Answer any five questions. Each question carries 4 marks.

1. Write and explain binary search algorithm.
 2. How does **bubble Sort** work? Sort the list 321, 45, 123, 98, 5 using bubble Sort.
 3. Write a recursive function to calculate the **GCD (Greatest Common Divisor)** of two numbers.
 4. Explain the process of **inorder, preorder, and postorder traversal** of a Binary Tree with an example.
 5. **Scenario-Based Question:**
An e-commerce website wants to sort thousands of product prices. Which sorting technique would you recommend between **Quick Sort** and **Merge Sort** for nearly uniform data, and why?
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Section B (3 Marks $\times$ 5 = 15 Marks)

1. What is the **base case** in recursion? Why is it essential?
 2. Write three differences between **Stack (LIFO)** and **Queue (FIFO)**.
 3. How is a **circular linked list** useful in real-world applications?
 4. List three real-life scenarios where **queues** are used.
 5. **Scenario-Based Question:**
You are designing a text editor where the user can undo/redo multiple actions. Which **data structure** would be most appropriate, and why?
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Section C (5 Marks $\times$ 3 = 15 Marks)

Answer any three questions. Each question carries 5 marks.

1. Write a program to create a **Binary Search Tree**, and implement a **search** function to find a given key.

2. Explain the **Selection Sort algorithm** and demonstrate its step-by-step working on the list: 9, 4, 7, 1, 6.

3. **Scenario-Based Question:**

A hospital is maintaining patient records in increasing order of admission time and allows fast insertion of new records and frequent searching. Suggest the most appropriate data structure or algorithm for this case and justify your answer.

Sample Question Paper – III

Subject: BCAC 0026 – Analysis of Algorithm and Data Structures**

Time: 3 Hours

Max Marks: 50

Section A (4 Marks × 5 = 20 Marks)

1. Explain the space and time complexity of Algorithms.
2. How is a **Binary Tree** different from a **Binary Search Tree (BST)**? Give examples.
3. Define **Recursion**. Write a recursive function to reverse a string.
4. **Scenario-Based Question:**
A video editing tool processes high-resolution frames and requires stable sorting by frame timestamp. Would you use **Merge Sort** or **Quick Sort**? Explain your choice with technical reasoning.
5. Compare the time and space complexities of **Bubble Sort** and **Insertion Sort**.

Section B (3 Marks × 5 = 15 Marks)

Answer any five questions. Each question carries 3 marks.

1. What is the difference between **tail recursion** and **head recursion**?
2. State any three operations possible in a **singly linked list**.
3. Explain **stack overflow** in recursive functions.
4. **Scenario-Based Question:**
You are designing a circular ticket counter display system that rotates names endlessly. Which data structure would you implement, and why?
5. Write a basic pseudocode for **Queue implementation** using arrays.

Section C (5 Marks × 3 = 15 Marks)

Answer any three questions. Each question carries 5 marks.

1. Design a program to implement a **Queue using a Linked List** and demonstrate enqueue and dequeue operations.
2. Perform all three traversals on a sample binary tree of at least 10 nodes.
3. **Scenario-Based Question:**
An airport manages flight queues dynamically and needs to efficiently serve the next flight based on urgency and priority. What data structure or algorithm would best serve this use case? Justify your answer with suitable reasoning.